

Definition and first terms

$5\,A(x) = 4 + 1\,x + A(x)^4$ , with the origin-normalized solution.  
 Normalization/grammar:  $A(x){=}1{+}(1){*}T(x)$ ;  $T{=}x{+}6{*}T^{^2}{+}4{*}T^{^3}{+}1{*}T^{^4}$ .  
 $a(0),\ldots,a(7) = 1, 1, 6, 76, 1201, 21252, 402892, 8001412$ .  
 Kernel used throughout:  $\rho(u) = u\,(1 - 6\,u^1 - 4\,u^2 - 1\,u^3)$ , so  $x = \rho(T(x))$ .

Typogeometry

literal 4-slot geometry; empty positions are shown. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities:  $\Delta_2$ : 6,  $\Delta_3$ : 4,  $\Delta_4$ : 1.

Coefficient and contour extraction

$$\mathcal{K}_n = \{(k_1,\ldots,k_3) \in \mathbb{Z}_{\geq 0}^3 : \sum_{j=1}^3 j k_j = n-1\}, \quad K = \sum_{j=1}^3 k_j,$$

$$a(n) = \frac{1}{n} \sum_{\mathbf{k} \in \mathcal{K}_n} \binom{n+K-1}{K} \binom{K}{k_1,\ldots,k_3} 6^{k_1} 4^{k_2} 1^{k_3}.$$

$$a(n) = \frac{1}{2\pi i n} \oint_{\gamma} \frac{du}{\rho(u)^n}, \quad n \geq 1.$$

Recurrence

$$\sum_{r=0}^3 P_r(n) a(n+r) = 0, \, n \geq 1, \text{ with}$$

$$P_0(n) = -256\,n^3 - 384\,n^2 - 48\,n + 40$$

$$P_1(n) = -3072\,n^3 - 9216\,n^2 - 8896\,n - 2752$$

$$P_2(n) = -12288\,n^3 - 55296\,n^2 - 79872\,n - 36864$$

$$P_3(n) = 491\,n^3 + 2946\,n^2 + 5401\,n + 2946$$

Differential equation

$$\begin{aligned} & (40\,x^3)\,A(x) + (-688\,x^4 - 2752\,x^3)\,A'(x) \\ & + (-1152\,x^5 - 9216\,x^4 - 18432\,x^3)\,A^{(2)}(x) + (-256\,x^6 - 3072\,x^5 - 12288\,x^4 + 491\,x^3)\,A^{(3)}(x) = 0. \end{aligned}$$

Matrices, telescoper certificate, and checks

$$U = \begin{pmatrix} \frac{11536}{491} & \frac{900}{491} & \frac{60}{491} & \frac{4}{491} \\ \frac{11040}{491} & \frac{736}{491} & \frac{180}{491} & \frac{12}{491} \\ \frac{3600}{491} & \frac{240}{491} & \frac{16}{491} & \frac{132}{491} \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad V = \begin{pmatrix} -1 & 0 & 0 & 0 \\ \frac{5644}{491} & \frac{409}{491} & \frac{60}{491} & \frac{4}{491} \\ \frac{3660}{491} & \frac{244}{491} & \frac{49}{491} & \frac{36}{491} \\ \frac{900}{491} & \frac{60}{491} & \frac{4}{491} & \frac{33}{491} \end{pmatrix}, \quad J = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Certificate.

$R(n,u) = N(n,u)/\rho(u)^2$ ,  $\deg_u N = 9$ .  
 Telescoping identity:  $\sum_r P_r(n)H_{n+r}(u) = \partial_u(R(n,u)H_n(u))$ .  
 Matrix dimensions:  $8 \times 8$ ; remainder  $3 \times 4$ , rank 3, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.